

Capstone failure testing

Project: Grounded Learning Service

1. Your mission

Your learning target

I can explain the mechanism and terminology of capstone failure testing.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of capstone failure testing

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Capstone service implementation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain capstone failure testing using the deciding step in this fictional service.

2. Notice the evidence

Capstone failure testing checks denied roles, missing evidence, invalid input, timeouts and stale data. Each expected outcome is written before the run.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Capstone failure testing — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

Check or calculation: Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Conclusion and limit: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Make retrieval unavailable as well.

Prediction: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase evidence but cannot replace missing evidence.

5. Words that unlock the idea

Word · Meaning

Word: Failure matrix

Meaning: Conditions with expected results and forbidden operations

Word: Trace assertion

Meaning: A check on actual operations

Word: Regression fixture

Meaning: A preserved case rechecked after change

Use two terms to explain a step in this week's example.

6. Follow a worked example

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone failure testing

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

The facts we will use

F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

Capstone failure testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Verify source

Retrieval supplies F4 and the time 15:00.



2. Inject failure

Disable drafting without changing source evidence.



3. Use bounded fallback

Return the checked template citing F4 and record degraded operation.

CHANGE ONE THING

Make retrieval unavailable as well.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Verify source

What happens and why: Retrieval supplies F4 and the time 15:00.

Step: 2. Inject failure

What happens and why: Disable drafting without changing source evidence.

Step: 3. Use bounded fallback

What happens and why: Return the checked template citing F4 and record degraded operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain evidence checks and honest status.

What this example does not establish

This trial covers one failure combination; check other components and interactions separately.

Read the labelled picture: Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that evidence gap?

Change one thing: Make retrieval unavailable as well. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

A case passes only all its declared response and trace requirements, not merely a convenient status code.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone failure testing”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that evidence gap?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

For Grounded Learning Service, use this week’s independent task as a rehearsal for the final evidence pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher’s fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Capstone failure testing — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

Check or calculation: Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Conclusion and limit: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of capstone failure testing

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of capstone failure testing

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the checked template citing F4 and record degraded operation.

- 1. Verify source
- 2. Use bounded fallback
- 3. Inject failure

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Use bounded fallback Return the checked template citing F4 and record degraded operation. Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain evidence checks and honest status.

Practice 2: explain the deciding evidence

Changed-case practice: Make retrieval unavailable as well. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Return unsupported or unavailable; do not use the factual template.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Return unsupported or unavailable; do not use the factual template. A template can phrase evidence but cannot replace missing evidence.

Repair a missing building block

Review Capstone service implementation (ai-6/33). Goal: Explain the mechanism and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Red team exercises (ai-6/31). Goal: Explain the mechanism and terminology of red team

exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction. Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Accessibility and user feedback (ai-6/28). Goal: Explain the mechanism and terminology of accessibility and user feedback. Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.